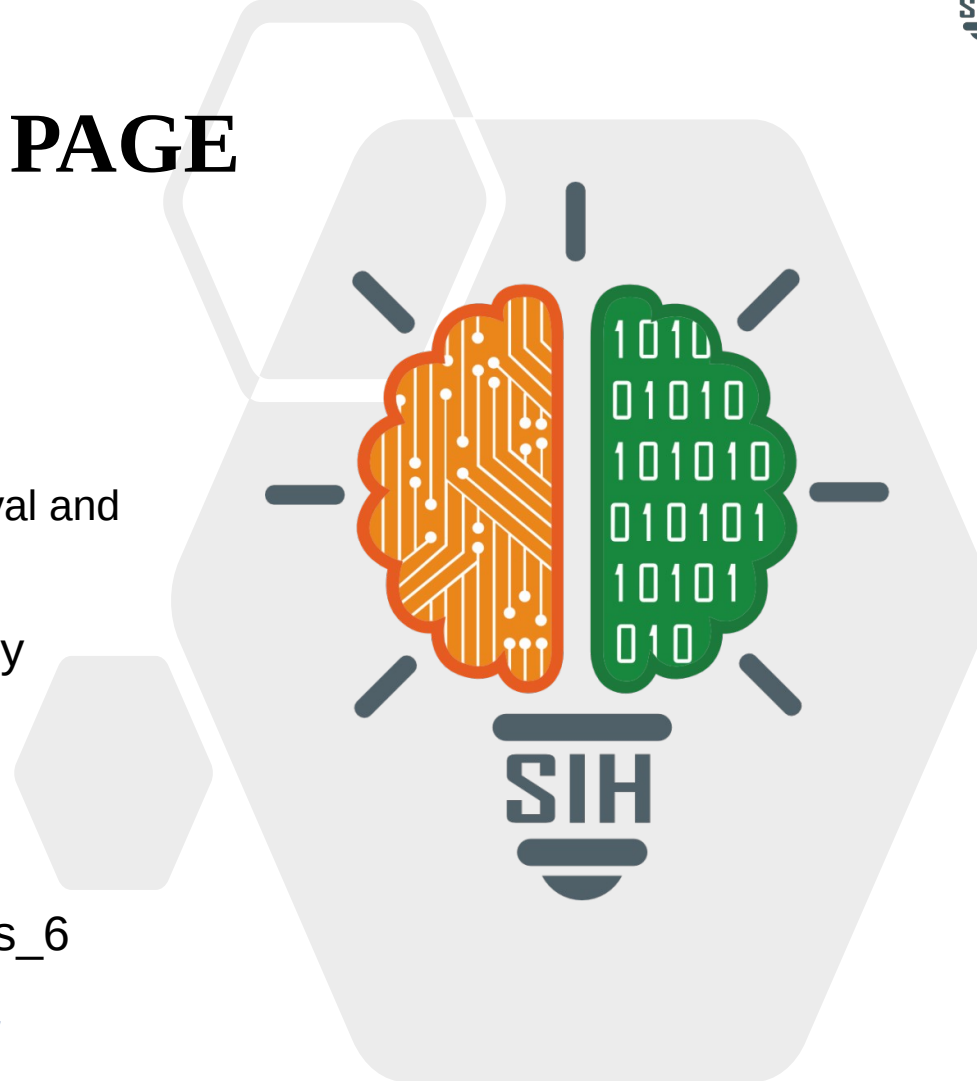




TITLE PAGE

- **Solution** – JalSrot
- **Problem Statement ID** – SIH26240
- **Problem Statement Title** – AI-Based Spring Revival and Recharge Planning for Tribal Areas
- **Theme** – Agriculture, Food and Rural Technology
- **PS Category** – Software
- **Team ID** – 178601
- **Team Name (Registered on portal)** – TechDevs_6

From scattered geospatial data to field-verified spring revival decisions.



1 Hidden recharge pathways

Surface terrain alone cannot reliably show where the groundwater feeding a spring originates.

2 Fragmented evidence

DEM, geology, rainfall, land use and spring observations sit in different datasets and formats.

3 Expert bottleneck

Remote tribal springs need hydrogeological expertise and field visits to assess.

4 Unsuitable interventions

Poorly chosen sites waste resources and can create environmental and geotechnical risk.

TODAY

Field survey

Expert interpretation

Site selection

Slow

Expensive

Difficult to scale

Limited feedback

JALSROT

Large-area computational screening

Candidate ranking

Expert verification

Narrowed search area

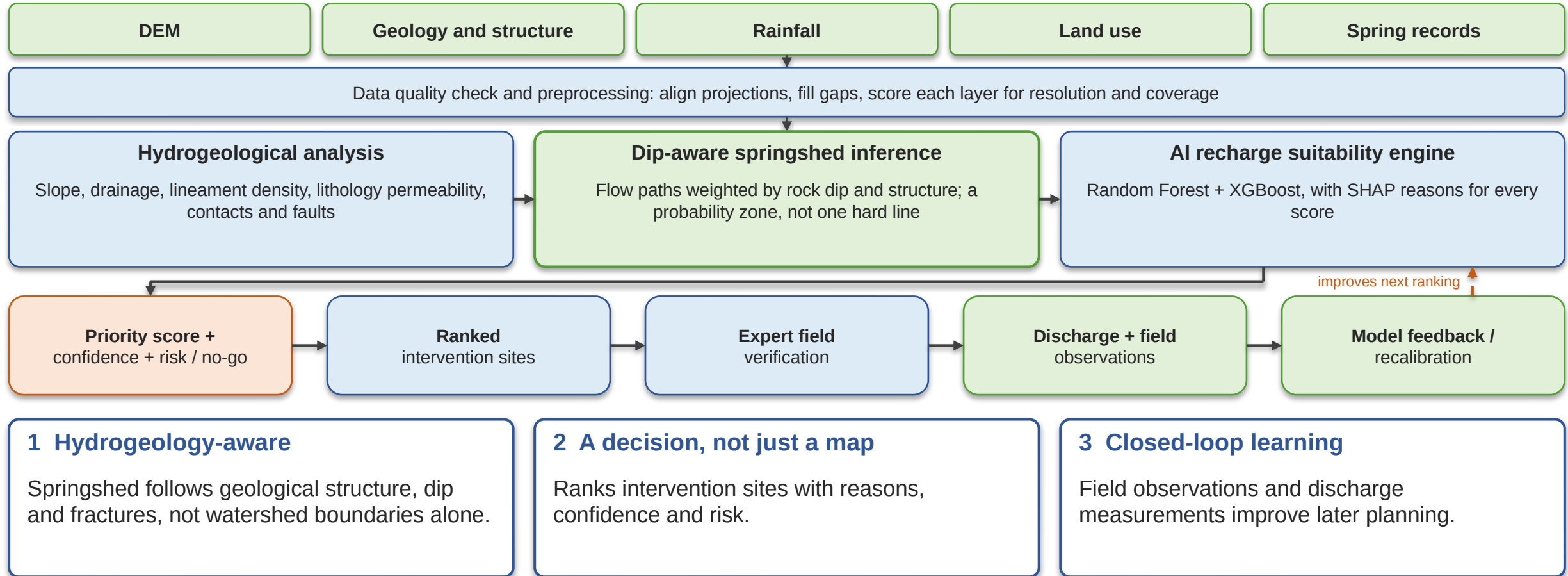
Ranked, with reasons

Confidence shown

Field feedback loop

JalSrot does not replace the hydrogeologist; it reduces the amount of terrain the hydrogeologist needs to inspect.

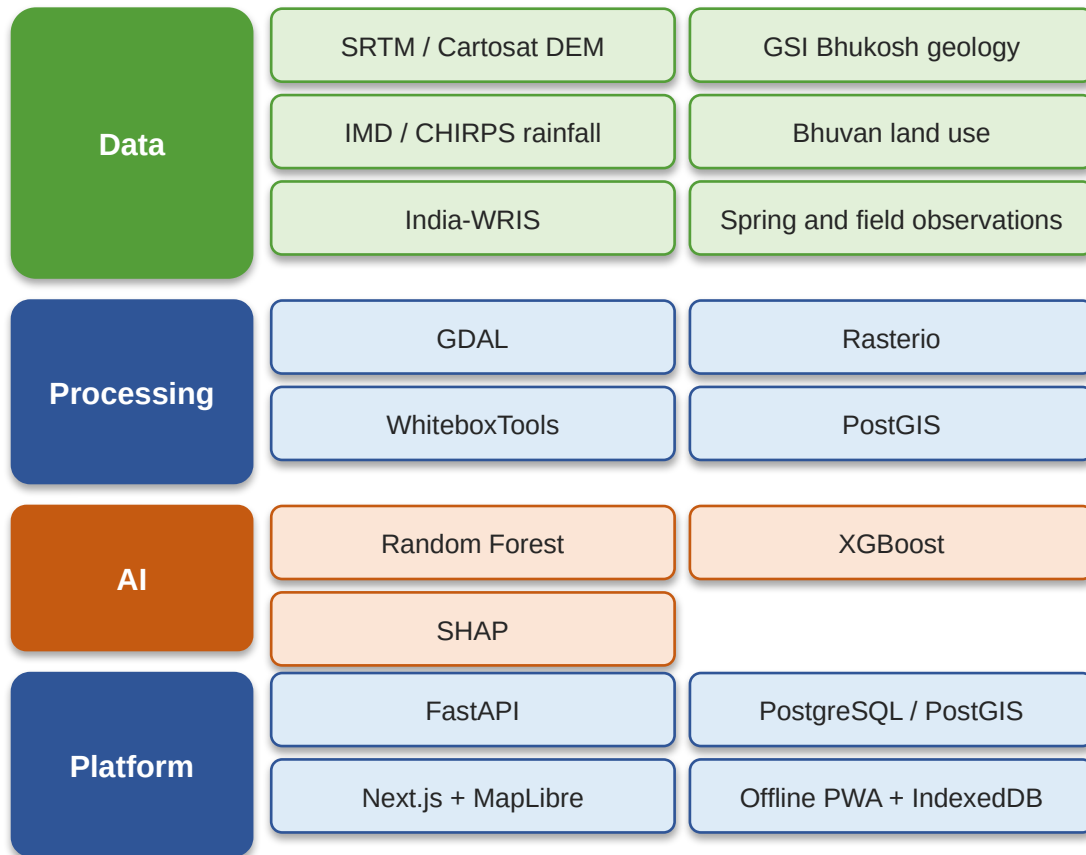
Why not just a GIS map? A map shows layers and leaves the judgement to the viewer. JalSrot also ranks candidate sites, states how confident it is, and flags where an intervention may not work or may be unsafe.



Physics sets the bounds, ML ranks within them. Geological and terrain constraints limit what the model may propose; ML then ranks the candidates that remain. AI does not replace hydrogeological reasoning.

Confidence reflects data availability and quality, and SHAP shows which factors drove each score.

Implementation



Proposed stack. Nothing here has been built or benchmarked yet.

How We Will Validate

- Historical validation**
Compare predictions with known spring behaviour and any historical observations.
- Geographic holdout**
Test on areas the model has not seen, to avoid spatial leakage.
- Expert validation**
Hydrogeologists review the highest-ranked candidate sites.
- Field validation**
Measure discharge and observations across wet and dry seasons.

Proposed metrics

Precision@K

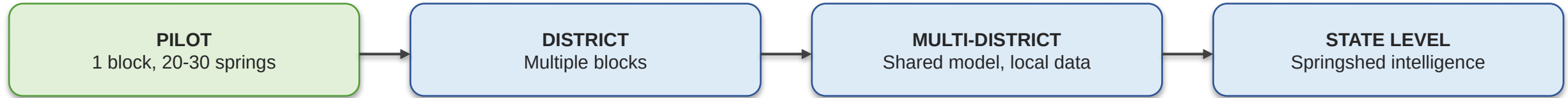
Recall

Ranking quality

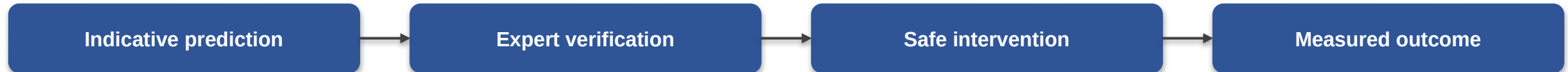
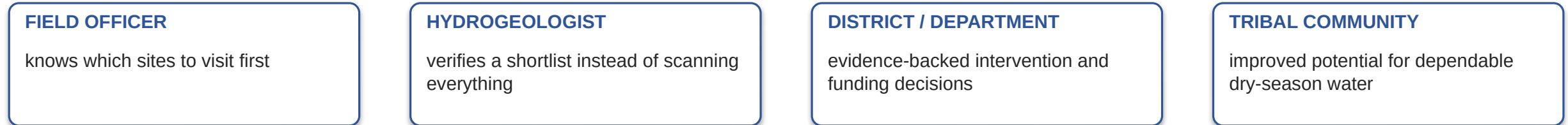
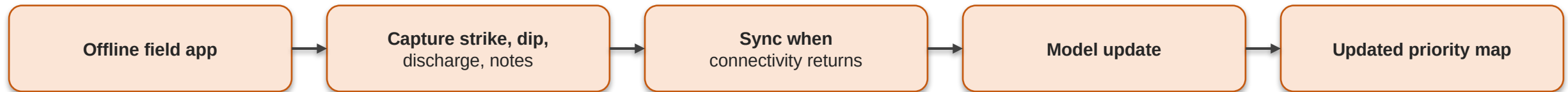
Confidence calibration

AI does not replace the hydrogeologist; it reduces the area they need to inspect.

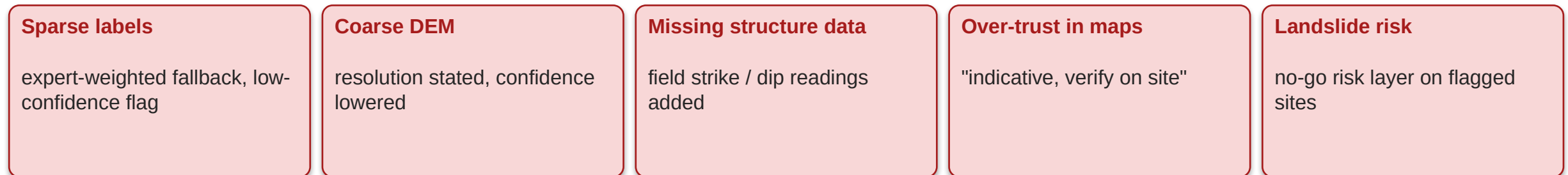
From Pilot Springshed to Scalable Field Deployment



Field loop (conceptual workflow)



Main risks and how they are handled



RESEARCH AND REFERENCES

S. No	Reference	Published by	Used for
1	SRTM Digital Elevation Model	NASA / USGS	Slope, aspect, drainage, flow paths
2	Bhuvan / Cartosat	ISRO – NRSC	Land use and finer terrain
3	Bhukosh	Geological Survey of India	Rock types, contacts, faults
4	CHIRPS rainfall dataset	UCSB Climate Hazards Center	Rainfall trends
5	India-WRIS	Ministry of Jal Shakti	Drainage and groundwater context
6	Inventory and Revival of Springs in the Himalayas	NITI Aayog	Springshed approach, revival practice
7	Random Forests (Breiman)	Machine Learning journal	Recharge suitability model
8	A Unified Approach to Interpreting Model Predictions (Lundberg & Lee)	NeurIPS	Explaining each site score
9	WhiteboxTools	University of Guelph	Terrain and hydrological analysis